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B. E. (Eighth Semester) Examination, April-May 2016

(New Scheme)

(Mech. Branch)

FINITE ELEMENT METHODS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. In question no. 1, 2 and 4 part (a) is compulsory. Attempt any two from parts (b), (c) & (d). In question nos. 3 and 5 part (a) is compulsory. Attempt any one from parts (b) and (c). Draw neat sketches, label it properly. Assume suitable data, if missing. Be precise in your answer.

1. (a) Name two commonly used approaches in deriving the finite element equations. 2

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(b) Obtain a two-term Galerkin solution for the problem : 7

$$\frac{d^2y}{dx^2} - 8x^2 = 4 \text{ in } 0 \le x \le 1$$

subject to the boundary conditions :

$y(0) = 0$ and $y(1) = 0$.

(c) Figure 1 (c) below shows an assembly of two bar elements made of steel and aluminium. Find the nodal displacements, element stresses, and the reaction force. Take : 7

E (for steel) = 210 GPa, Area, A (for steel) = 2 cm^2,

E (for Al) = 70 GPa, Area, A (for Al) = 4 cm^2.

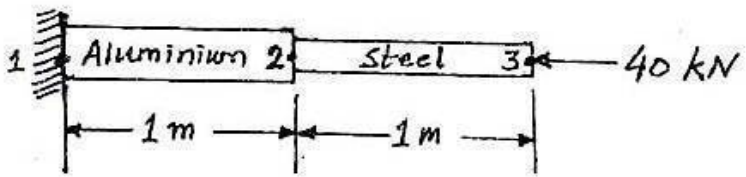

fig.1(c)

(d) A system of four springs is shown in the figure 1 (d). Apply the principle of minimum potential energy to

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obtain the Global Stiffness Matrix. The vertical members at nodes 2 and 3 are to be considered rigid. Solve and obtain : 7

(i) Nodal displacements at 2, 3 and 4

(ii) Reaction force at node 1.

Take :

$K_1 = 2$ N/mm $\quad F_2 = 30$ N

$K_2 = 4$ N/mm $\quad F_3 = 0$

$K_3 = 3$ N/mm $\quad F_4 = 40$ N

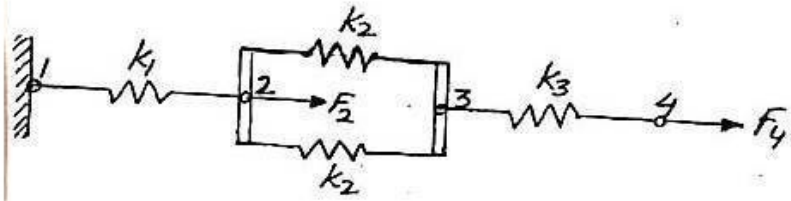

fig.1(d)

(a) What are the three types of loading considered in structural mechanics problem? 2

(b) Derive the shape functions of a linear, elastic bar element having two nodes. 7

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(c) Derive the shape functions of a quadratic, elastic bar element having three nodes. 7

(d) Consider the bar in the figure shown in fig. 2 (d). 7

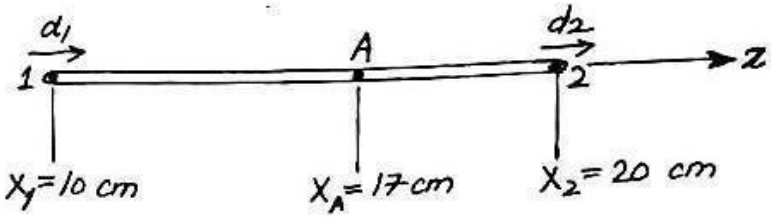

fig.2 (d)

Bar cross-sectional area, $A = 1{\cdot}4$ cm^2, Young's modulus, $E = 2 \times 10^{11}$ N/m^2. If $d_1 = 0{\cdot}02$ cm and $d_2 = 0{\cdot}01$ cm, determine the following :

(i) the displacement at point A

(ii) the strain in the element

(iii) the stress in the element

(iv) the element stiffness matrix

3. (a) What is the difference between bar, beam and frame element? Be precise in your answer. 2

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(b) Derive the stiffness matrix for the beam element. 14

(c) Derive the stiffness matrix for the frame element. 14

4. (a) Why the three-node triangular element is also known as Constant Strain Triangular (CST) element? 2

(b) Derive the shape functions of a four-noded quadrilateral element using serendipity approach. 7

(c) Use Gaussian Quadrature to obtain exact values of the following integrals. 7

(i) $\int_{-1}^{1} \left(3x^3 + x\right) dx$

(ii) $\int_{0}^{3} \left(x^2 - 1\right) dx$

(d) For the three-noded triangular element shown in fig. 4 (d), calculate the temperature at point $P(3, 2 \cdot 5)$. The nodal temperatures are : 7

$T_1 = 100$ °C

$T_2 = 300$ °C

$T_3 = 400$ °C

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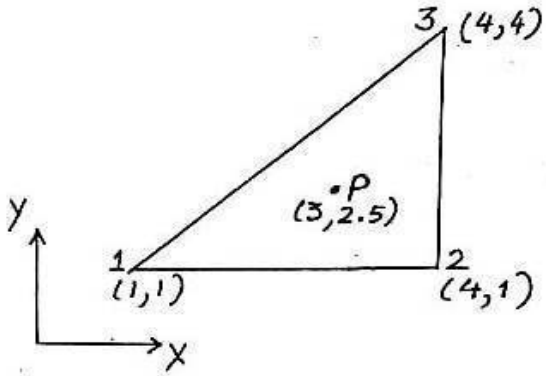

fig.4 (d)

5. (a) Write the 3×3 constitutive material property matrix for plane strain problem. 2

(b) Derive the strain-displacement relationship matrix for three-noded triangular element in plane stress. 14

(c) Assuming plane stress conditions, determine the element stresses σ_x, σ_y and τ_{xy}, for the element shown in fig. 5 (c). The nodal displacements are :

$u_1 = 2 \cdot 0$ mm $\quad v_1 = 1 \cdot 0$ mm

$u_2 = 0 \cdot 5$ mm $\quad v_2 = 0 \cdot 0$ mm

$u_3 = 3 \cdot 0$ mm $\quad v_3 = 1 \cdot 0$ mm

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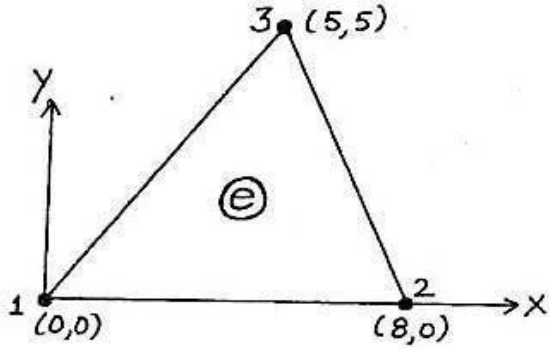
Take :

$E = 210$ GPa

$\upsilon = 0{\cdot}25$

Thickness $t = 2$ mm. 14



All coordinates are in cm

fig.5 (c)

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